UNIVERSITY AT ALBANY
State University of New York

College of Arts and Sciences
Department of Mathematics and Statistics

# Algebra/Topology Seminar

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## Complex Hyperbolic Gromov–Thurston Metrics

Thursday, November 19, 2026
3:00 p.m. in Massry B008

Abstract. In 1987 Gromov and Thurston discovered the first Riemannian manifolds that are not homotopy equivalent to a hyperbolic manifold, but admit a Riemannian metric whose sectional curvatures are pinched arbitrarily close to $-1$. The manifolds that they construct are branched covers of hyperbolic manifolds, and to construct the metric they perform a sort of "geometric surgery" about the ramification locus. In 2022 Stover and Toledo proved the existence of similar branched cover manifolds built out of complex hyperbolic manifolds, and via a result of Zheng these manifolds admit a negatively curved Kahler metric. In this talk we will discuss how to construct a (not Kahler) Riemannian metric on these Stover–Toledo manifolds which is arbitrarily close to being negatively 1/4-pinched. This provides the first known examples of Kahler manifolds that are not homotopy equivalent to a complex hyperbolic manifold, but admit a Riemannian metric with this property.